INSTITUTE OF DISTANCE LEARNING

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AE 464

CROP STORAGE AND STORAGE STRUCTURES

Lecture

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STORAGE OF PERISHABLE CROPS

Introduction

- Perishable crops are crops with moisture content above 60%. Examples of perishable crops are cassava, tomatoes and oranges.

- Due to their high moisture content, perishables are expensive to store and very perishable under ambient conditions.

- The aim of storage is to provide an environment which slow down the rate at which decay processes occur.

PRE-COOLING FOR STORAGE

- A great deal of agricultural produce will be stored for lengthy periods before consumed.
- Such produce is, by definition, biological in nature, and will therefore decay over time.
- The aim of storage is therefore to provide an environment which slows down the rate at which these processes occur.
- This is usually done by cooling the produce, and by controlling the humidity and composition of the atmosphere in the store.

PRE-COOLING FOR STORAGE

- Crops are cooled in cooling rooms, where refrigeration systems are set up to cool air which is then forced through the produce.

- The air is cooled by a heat exchanger through which the coolant flows, and this is forced through the ducting and into the fruit by the fan.

- The covers and the configuration of the pallets must be designed so that all the fruits receive a similar flow of cooling air.

PRE-COOLING FOR STORAGE

- Another method of cooling is known as hydro-cooling. Instead of air the crop is immersed in cooled water.

- As water has a high specific heat, and because a higher heat transfer coefficient between the water and the crop is possible, hydro-cooling will extract heat from a crop very quickly, so that greater quantities of fruit can be handled in a given time.

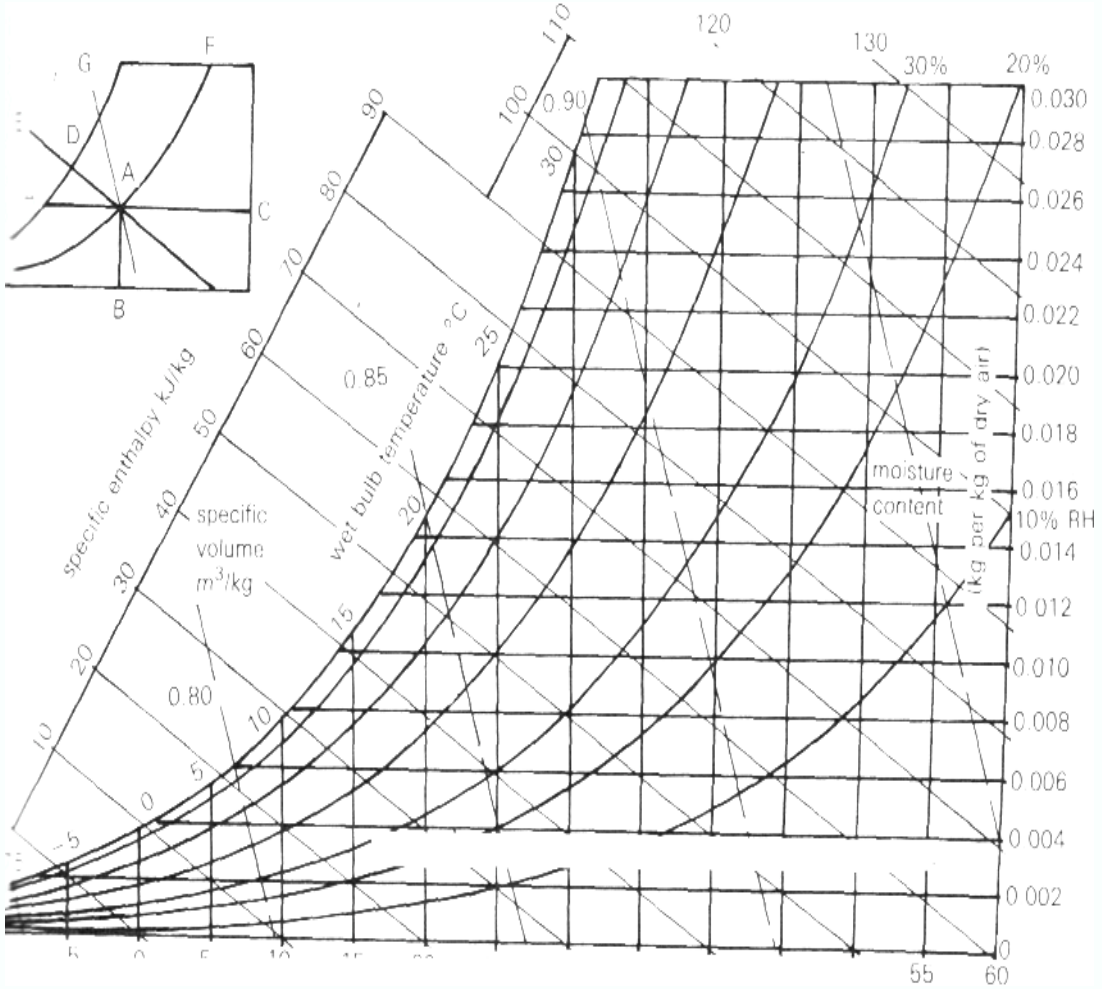
- Vacuum cooling can be used to cool leafy crops such as lettuce, while some crops such as broccoli can be cooled and stored by packing them in crushed ice.

COOLING CAPACITY CALCULATIONS

- Fruit must be cooled as quickly as is economically possible. In order to do this the capacity of the cooling plant needs to be calculated.

- This requires a knowledge of the internal energy which will need to be removed in the form of heat (in most of the terminology the term heat is used to describe this internal energy, and we even refer to the specific heat to quantify it.

COOLING CAPACITY CALCULATIONS



METHODS OF COOLING

Products may be cooled by means of:

-Cold air

- Room cooling
- Forced air cooling

-Cold water

- Contact icing
- Hydro cooling

-Vacuum cooling

-Evaporation cooling

Room cooling

- The most commonly used technique.
- The produce are in boxes, cartons, bulk containers and other packages exposed to cold air in a normal cool store.
- The product is cooled and stored at the same time. There is a refrigeration unit attached to the store.

Forced air cooling

- Forced air cooling can cool the product in a shorter time than room air cooling since room cooling only removes heat from the surface of the product whiles forced air cooling removes heat from both the surface and within of the product.

- Example of forced air cooing is tunnel cooling.

- In tunnel cooling, either the air moves or the product is moved by means of trolleys, conveyors. This is practised extensively in South Africa.

Hydro cooling

- In hydro cooling, water acts as heat transfer medium.
- The principle behind the use of this method is that the heat capacity of water is far greater than that of air. Produce are passed under cold showers in a moving conveyor.
- The product is cooled and clean. In air cooling, the product loses little weight but in hydro-cooling the weight of the product is managed.

Contact Icing

- Ice is crushed and the product placed in it or the ice is placed on the product.

Vacuum Cooling

- It is a special technique used for vegetables. It is uniform cooing at low pressure.
- The product is loaded into a sealed container and the pressure is reduced considerably.
- At this pressure, the produce is cooled by evaporation of water from the surface of he produce.

Evaporative cooling

- It is a technique by which dry air is cooled by blowing it across a wet surface. It is effective in dry climates (areas where relative humidity is low).

- Evaporative cooling has low energy cost.

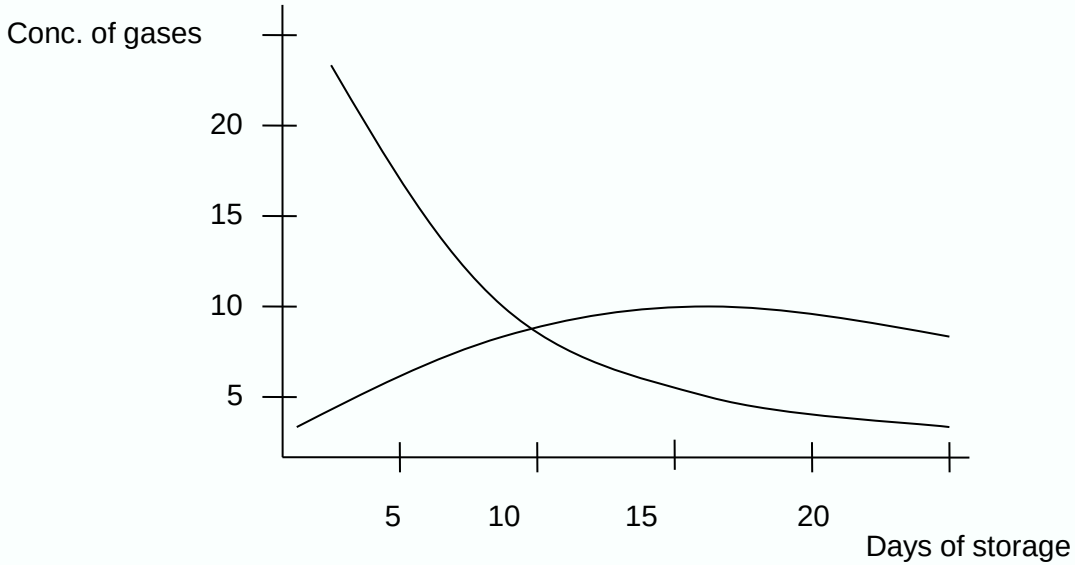
- Two methods are used:

-Humidity cooled air for cooling.

-Misting; spraying water into air.

CONTROL ATMOSPHERE STORAGE

- It involves modification of the storage environment by limiting oxygen and increasing the carbon dioxide concentration in the storage atmosphere.



- The oxygen content is reduced below 10% whiles the carbon dioxide concentration is increased to about 4 – 5%.

COOLING CAPACITY OF REFRIGERATION PLANTS

- **Product load (Field heat)**
- Product load: it is the load energy which must be removed to reduce the temperature of the produce to the desired storage temperature.
- The product load, Hp is determined as:
- $Hp = \frac{2.08MS(T_f - T)}{t_s}$

- Hp– watts (W), t_s – Time(s), M– Mass of produce/crop (kg),

S – Specific heat capacity of the produce J/kg^{0}C,

T – Refrigeration temperature, T_f –Fruit /crop temperature

COOLING CAPACITY OF REFRIGERATION PLANTS

QUESTION

- Crop at a temperature of 25C is to be precooled to 0^0C in 12 hours. The heat capacity of the product is 3650J/kg^{0}C. The weight of the product is 15tons.

- Determine the product load.

RESPIRATION LOAD

- The crop is a living breathing product and gives off heat as it breathe.
- It depends on the metabolic activity and also temperature.
- The respiration load or respiration heat is given by
- Hr = MH
- M – Product mass (kg), H – Metabolic heat/ respiration heat/ heat given off (W/kg)
- For pre-cooler design the maximum value, which will beat the initial crop temperature Tf, should be chosen.

RESPIRATION LOAD

Question

- The metabolic heat at O^0C for ripe tomatoes is 0.05, 0.009 for onions, 0.012 for apples, and 0.009 for oranges.

- Given 10 tonnes of each product, determine which of the product will give the lowest respiration load.

WALL LOAD

- $Hw = UA\ (Tx - T)$

- Tx = Temperature of medium- Outside temperature (^0C),

T = Inside temperature- desired storage temperature (^0C),

Hw = Wall load (W),

U = Heat transfer co-efficient ($W/m^2\ ^0C$),

A = Total outside surface area of the cold store (m^2)

WALL LOAD

Question

- Given the dimension of a ware house to be 9m x 9m x 4m

- Calculate the total wall load for outside temperatures of 30°C, under floor temperatures of 20°C and a floor temperature of 0°C.The heat transfer of the various mediums is as given

- Walls – 0.241 $W/m^2°C$, Ceiling – 0.21 $W/m^2°C$,

Floor – 0.35 $W/m^2°C$

AIR CHANGE LOAD

- Due to

-air itself

-water component

- Ha =Ma (ha -h) + MwSw (Ta -T)

Where M_a is the mass of air admitted in kg per second

h_a is the enthalpy of the outside air (J/kg),

h is the enthalpy of the air at the conditions inside the cooler, M_w is the mass of water which condenses out (kg/s), and

- S_w is the specific heat of water.

- The enthalpy and the mass of water are determined from the psychometric chart.
- This is a chart which gives us the properties of moist air at atmospheric pressure, providing we know the relative humidity and the temperature.

OTHER LOADS

- We allow for other heat loads produced by fans, motors, people, and lights. These loads are also hard to estimate and a sensible guess is appropriate.

EXAMPLE

- Calculate the size of a refrigeration plan required to cool 15 tonnes of apples to 0°C from 20°C in 12 hours, in a pre-cooler. Use the following data:
- Respiration rate for apples at 20 °C = 0.092 W/kg, U for cooler walls and ceiling = 0.241 W m^{-2} $°C^{-1}$, out temperature 29.5 °C, U for floor = 0.35, Underfloor temperature 14 °C, size of cooler 9mx9m by 4m high, humidity of out air = 70%, assume one air change per hour.

OTHER LOADS

Solution: First we will calculate the heat loads, and then find the minimum surface conductance coefficient required to extract the heat from the apples,

- a. The product load is given by:

-P_p = {2.08 x 15,000 x 3650 x (20 - 0)}/12 x 3600

= 52.73 kW

- b. The respiration load when the apples are first put in is:

-P_r = 15,000 x 0.092 = 1.38 kW When they have cooled to 0°C the load be 15,000 x 0.012 = 0.18 kW.

-Thus the load will drop as cooling proceeds, and in storage the lower value can be used. However for the pre-cooler, the greatest value is appropriate i.e., 1.38 kW.

OTHER LOADS

- c. For the ceiling, walls, and floor respectively, The heat load is:

-P_{w1} = 0.241 x 9 x9 x (29.5-0) = 576 W

-P_{w2} = 0.241 x 9 x 4 x 4 x (29.5 - 0) = *1024*

-P_{w3} = 0.35 x 9 x 9 x (14 - 0) = 397 W

The total wall gain is thus the sum of these or 1.997 kW.

- d. Calculation of the air change load requires the use of the psychrometric chart. First the volume of air inside the cooler can be found by finding the point representing the moist air at 0°C and 100% RH, and reading the value of the volume from the near vertical lines (i.e., lines parallel to AG in the inset).

-*At 0°C,* one kg of dry air occupies a volume of 0.778 m^3. Hence for an air change of 1, the mass of air is:

OTHER LOADS

-M= (9 x 9 x 4) /0.778= 416.5kg

-The enthalpy can be found. This is obtained by following the sloping lines back the sloping scale on the left (i.e., lines parallel to AE).

Thus, at 29.5 °G and 70% RH, the enthalpy h_a = 78.5 kJ/kg dry air, and at 0°C and 100% RH, h = 9.2 kJ/kg dry air. The mass of water per kg of dry air is found by following the horizontal lines to the right and reading off the scale at G.

This gives values 0.0189 kg and 0.0037 kg respectively, Hence the mass of water which must be condensed is:

-M = (0.0189 - 0.0037) x 416.5 = 6.33 kg.

-Thus for the total enthalpy change per second is given by:

-$M_a(h_a - h)$= {416.5 x (78,500 - 9,200)} /3600

=8.017 kW

OTHER LOADS

-For the water vapour:

-$M_w S_w (T_a - T) = \{6.33 \times 4{,}200 \times (29.5-0)\}$

$= 0.217$ kW only,

-$P_a = 8.017 + 0.217 = 8.234$ kW

-In this calculation, the second term, the load to cooling the condensed water, is only 217W.

• e. We will allow a value of 400 watts for the incidental load.

Thus the total heat load is given by adding these together. Thus:

Total heat load

-$P = 52.73 + 1.38 + 1.997 + -0.40$

$= 64.7$ kW

- As can be seen from these figures, the load is dominated by the field heat, so it is evident that a much larger cooler is required to cool the produce than to maintain it once cooled.
- Now we must ensure that the heat can be removed from the apples.
- This will be possible providing t_a is less than or equal to t_s. The minimum surface heat transfer coefficient can be calculated from *Equation 11.5*.
- Assuming a mass of 0.3 kg, and a surface area of 0.025 m^2 for an apple 90 mm in diameter, the minimum value of h is given by:

$$h_s = \frac{2.08MS}{\Delta t_s}$$

$$h_c = \frac{2.08MS}{At_c} = 2.1 \quad Wm^{-2}{}^{o}C^{-1}$$

- This is quite low, so providing the air can flow over the apples the required cooling time should be attainable
- Ma- mass of air admitted in kg per second, ha is the enthalpy of the outside air (J/kg), h is the enthalpy of the air at the conditions inside the cooler, Mw

CHILLING INJURY

- Some fresh produce are susceptible to injury at low but non-freezing temperature.
- Symptoms of chilling may not be developed until the produce is removed from cold storage to normal ambient temperature.
- Effect of Chilling Injury
- Discolouration- Internal or external

-Abnormal ripening; eg uneven ripening

-Increase in decay; activity by micro-organisms

-Skin piking; sunken spots

CHILLING INJURY

Commodities	Approximate lowest safe temp (^{0}C)	Chilling injury symptoms
Aubergine	7	
Avocado	5-13	
Banana (green/ripe)	12-14	
Cucumbers	7	
Beans (green)	7	
Grapefruit	10	
Lemons	13-15	
Mangoes	10-13	
Okra	7	
Tomatoes (mature green)	13	
Tomatoes (ripe)	7-10	

HANDLING DAMAGE

- There are three ways in which crops can be damaged mechanically, and different crops have different levels of susceptibility to these types of damage.

- Crops can be damaged by **impact**, by **compression** due to (for example) stacking or squeezing, or by **abrasion** due to rubbing or vibration.

- The resistance of produce to mechanical damage can be measured by a variety of tests, including compression and drop impact tests.

- Bruising can occur at any point in the harvesting and processing cycle.

HANDLING DAMAGE

- Another major source of damage can be the **grading machine**
- Some crops such as potatoes and mangos can be bruised internally by impact, even though the surface appears to be fine.
- **Once damaged, the storage life of the product can be seriously affected.**
- In some fruits the damage region decays rapidly, with the production of gases such as ethylene.
- Kiwifruit is an example of crop, as it is highly sensitive to ethylene:

STORAGE AFTER COOLING

- After cooling, the crop is stored in a cool store until required.

- For each crop advisors can detail the ideal post-harvest storage conditions.

- For example, a kiwifruit cool store should be sited well away from any main road, because the minute traces of ethylene in car engine exhaust fumes can raise the local air levels sufficiently to cause premature ripening.

STORAGE AFTER COOLING

- Once grain and cereal crops have been dried they must be stored and kept at low moisture levels. Ambient air can be used to do this.

- For example the equilibrium moisture content in grain is 14% at 65% relative humidity of the air, so providing the air is below 65% RH, it can be used to keep the grain dry.

- It is also necessary to ventilate the grain in order to remove heat produced by respiration of the grain.

STORAGE AFTER COOLING

- Bulk stores of produce can be very large. It is necessary to place sensors in appropriate positions throughout the produce, so that the actual behaviour of the environment can be sensed.

- The least costly form of storage involves adjusting the physical properties of the air such as temperature and relative humidity.

- It is difficult to measure high relative humidities (above 95%) accurately, and it is not difficult air to become saturated so that water condenses on the crop, and this can harmful.

Thank You

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